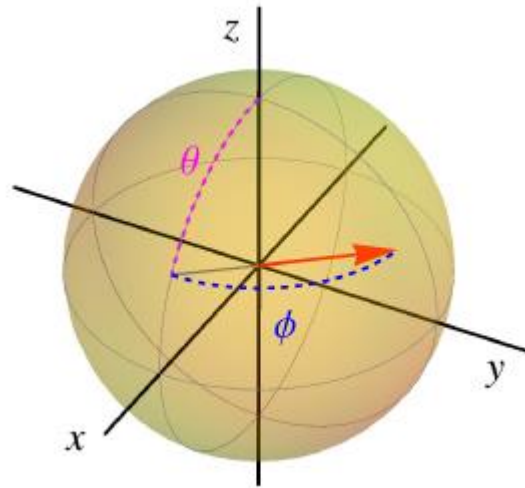


Quantum Computation

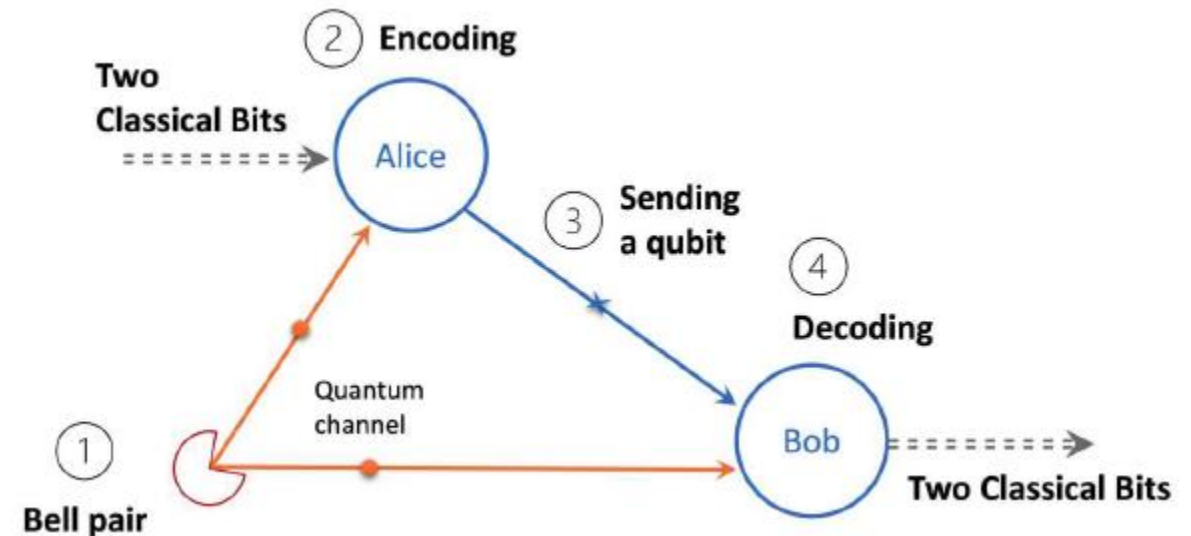
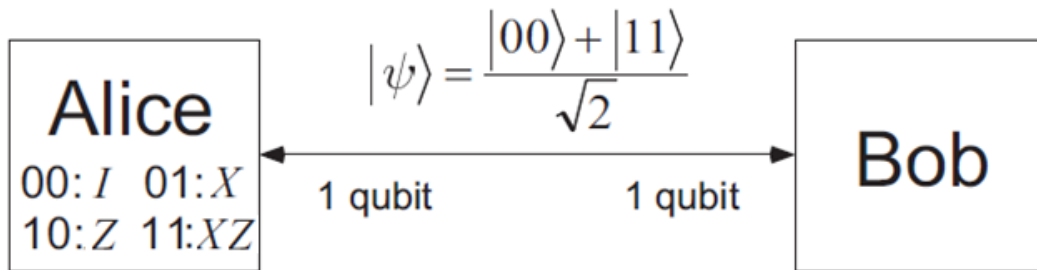
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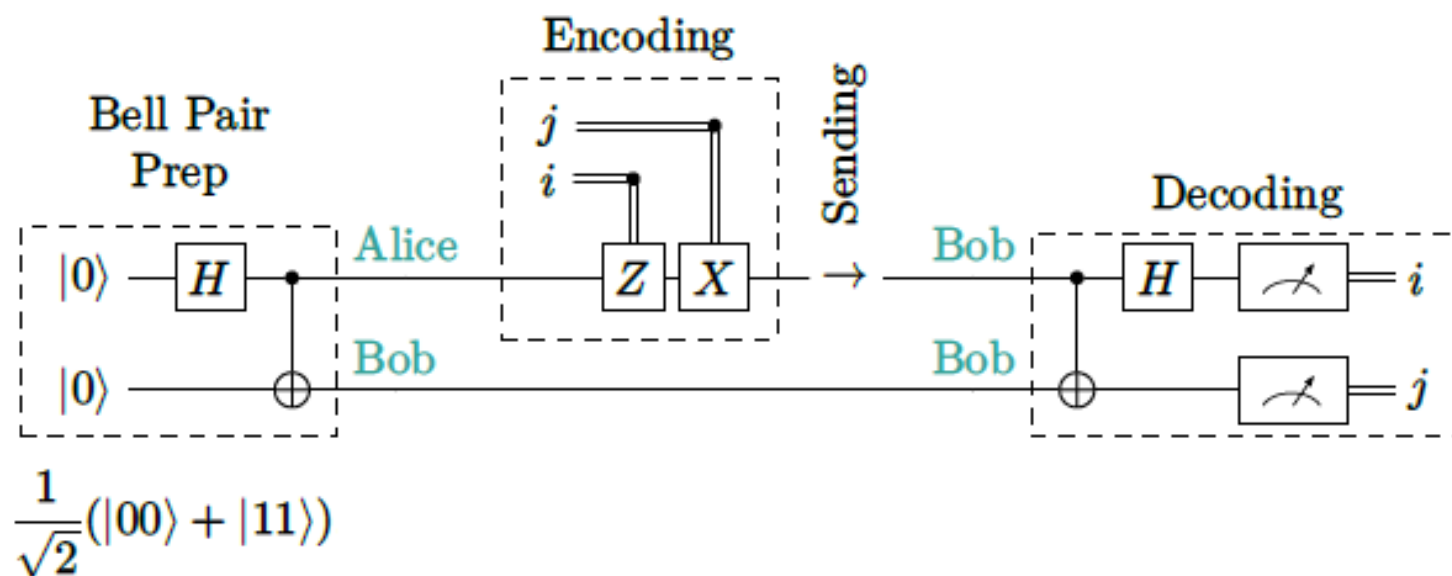
Week #7



Superdense coding

Superdense coding is a quantum communication protocol that allows someone to send two bits of information using just one quantum bit (qubit). It's one of the fascinating advantages of quantum entanglement.

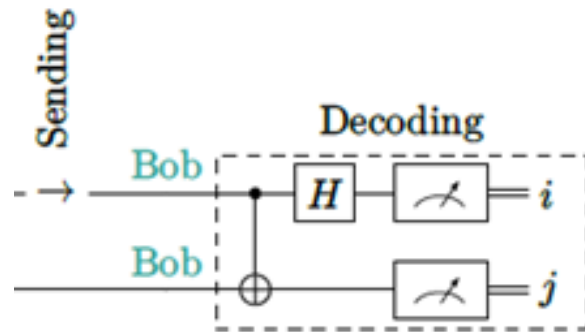




Bits to Send (ij)	Encoding Gates	Encoded State (New Bell State)
00	I (identity)	$ \Phi^+\rangle \equiv \beta_{00}\rangle = \frac{1}{\sqrt{2}}(00\rangle + 11\rangle)$
01	X (bit-flip)	$ \Psi^+\rangle \equiv \beta_{01}\rangle = \frac{1}{\sqrt{2}}(10\rangle + 01\rangle)$
10	Z (phase-flip)	$ \Phi^-\rangle \equiv \beta_{10}\rangle = \frac{1}{\sqrt{2}}(00\rangle - 11\rangle)$
11	X & Z or iY	$ \Psi^-\rangle \equiv \beta_{11}\rangle = \frac{1}{\sqrt{2}}(01\rangle - 10\rangle)$

$$|\beta_{00}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \xrightarrow{1\otimes 1} \text{Then sends her qubit to BOB}$$

Assume Alice wants to send 00:



$$\text{BOB: } \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \xrightarrow{CNOT} \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$$

$$\xrightarrow{H\otimes 1} \frac{1}{2}[(|0\rangle + |1\rangle)|0\rangle + (|0\rangle - |1\rangle)|0\rangle]$$

$$= \frac{1}{2}[(|0\rangle + |1\rangle + |0\rangle - |1\rangle)|0\rangle]$$

$$= |0\rangle|0\rangle$$

Phase kickback and probe qubit

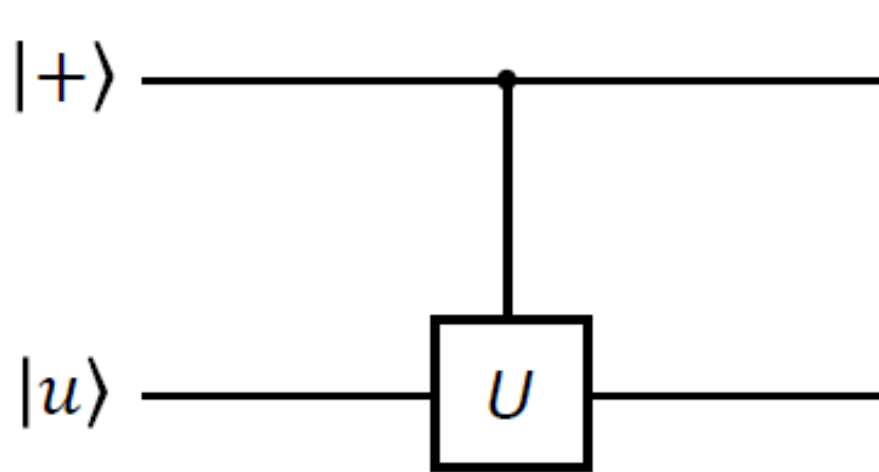
The eigenvalues of unitary operators are phases.

$$U|\psi_\lambda\rangle = e^{+i\lambda}|\psi_\lambda\rangle \quad \lambda = 2\pi\varphi \quad 0 \leq \varphi \leq 1 \quad \varphi = 0.x_1x_2x_3 \dots$$

$$e^{2\pi i\varphi} = e^{2\pi i0.x_1x_2x_3}$$

$$e^{2\pi i2^1\varphi} = e^{2\pi ix_1.x_2x_3} = \underbrace{e^{2\pi ix_1}}_1 e^{2\pi i0.x_2x_3}$$

- Now, assume: $U|u\rangle = e^{2\pi i\phi}|u\rangle$



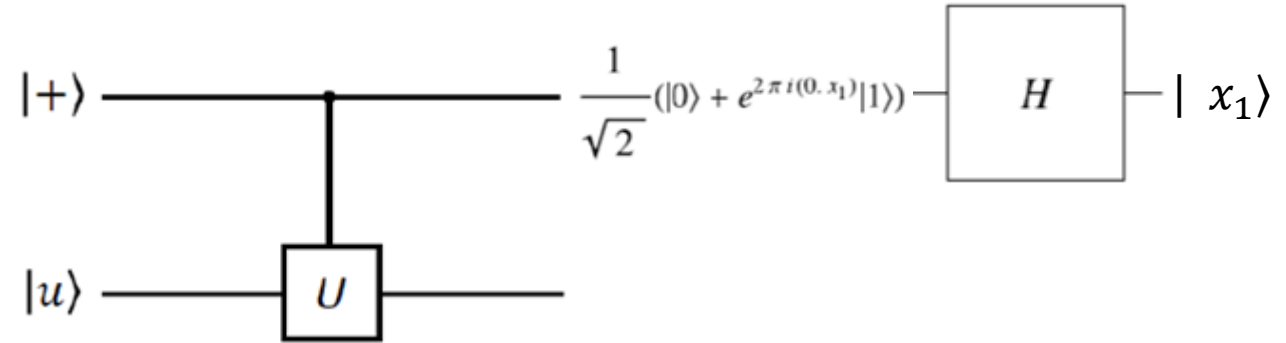
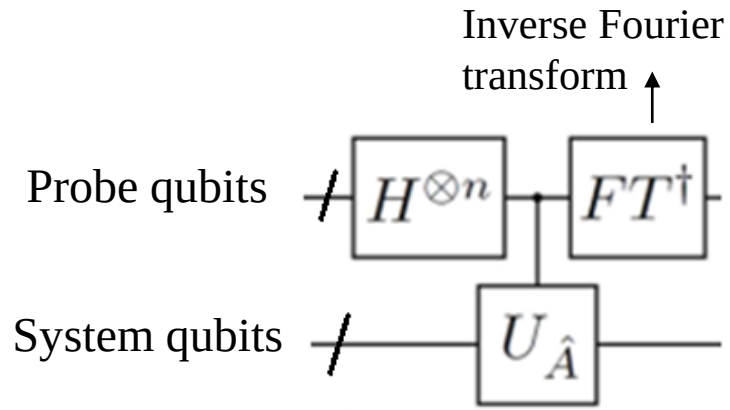
$$\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|u\rangle$$

$$\rightarrow \frac{1}{\sqrt{2}}(|0\rangle|u\rangle + |1\rangle e^{2\pi i\phi}|u\rangle)$$

$$= \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i\phi}|1\rangle)|u\rangle$$

But How to probe ϕ ?

Single qubit example: Hadamard transform is the single qubit FT!



Probe qubit: $\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 0.x_1}|1\rangle) \xrightarrow{H} \frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) + \frac{e^{2\pi i 0.x_1}}{\sqrt{2}}(|0\rangle - |1\rangle)\right)$

$$= \frac{1}{2}\left((|0\rangle + |1\rangle) + e^{2\pi i 2^{-1}x_1}(|0\rangle - |1\rangle)\right) = \frac{1}{2}\left((|0\rangle + |1\rangle) + e^{i\pi x_1}(|0\rangle - |1\rangle)\right), (e^{i\pi} = -1)$$

$$\begin{cases} = |0\rangle, & x_1 = 0 \\ = |1\rangle, & x_1 = 1 \end{cases}$$

$$= |x_1\rangle$$

Reminder

Hadamard $\text{---} \boxed{H} \text{---}$ $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

$$R_k \equiv \begin{bmatrix} 1 & 0 \\ 0 & e^{2\pi i/2^k} \end{bmatrix}$$

Phase $\text{---} \boxed{S} \text{---}$ $\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$

$$R_2 = S, R_3 = T$$

$\pi/8$ $\text{---} \boxed{T} \text{---}$ $\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$

Quantum Fourier transform

One of the most useful ways of solving a problem in mathematics or computer science is to *transform* it into some other problem for which a solution is known.

One such transformation is the *discrete Fourier transform*.

The discrete Fourier transform takes as input a vector of complex numbers, $x_0, \dots, x_{N-1}$ outputs the transformed data, a vector of complex numbers $y_0, \dots, y_{N-1}$, defined by

$$y_k \equiv \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} x_j e^{2\pi i j k / N} .$$

The *quantum Fourier transform* on an orthonormal basis $|0\rangle, \dots, |N-1\rangle$ is defined to be a linear operator with the following action on the basis states,

$$|j\rangle \longrightarrow \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{2\pi i j k / N} |k\rangle . \quad (46)$$

Here, i is the complex number, and k is a number in base 2.

This is not a very descriptive formula , let's define the items;

For example:

$$(423)_{10} = 10^3 \left(\frac{4}{10} + \frac{2}{10^2} + \frac{3}{10^3} \right) = 10^3(0.4 + 0.02 + 0.003)$$

$$J = (J_1 J_2 J_3)_2 = 2^3 \left(\frac{J_1}{2} + \frac{J_2}{2^2} + \frac{J_3}{2^3} \right) = 2^3(0.J_1 + 0.0J_2 + 0.00J_3)$$

Therefore in general, we define the QFT coefficients as

$$e^{2\pi i 0.J_1 J_2 \dots J_n} = e^{2\pi i \frac{J}{2^n}}$$

Here, i is the complex number, and k is a number in base 2.

n is the number of qubits and $N = 2^n$ is the number of eigenstates.

$$\begin{aligned}\text{QFT}_{2^n} |j\rangle &= \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} e^{2\pi i \frac{jk}{2^n}} |k\rangle \\ &= \frac{1}{\sqrt{2^n}} \sum_{k_1=0}^1 \sum_{k_2=0}^1 \cdots \sum_{k_n=0}^1 e^{2\pi i j (\sum_{\ell=1}^n k_\ell 2^{-\ell})} |k_1 k_2 \dots k_n\rangle\end{aligned}$$

For instance, for the case of $n = 2$,

$$\begin{aligned}
 QFT|j\rangle &= \frac{1}{\sqrt{2^2}} \left(\sum_{k_1=0}^1 \sum_{k_2=0}^1 e^{2\pi i j (\sum_{l=1}^n k_l 2^{-l})} |k_1 k_2\rangle \right) \\
 &= \frac{1}{\sqrt{2^2}} \left(\sum_{k_1=0}^1 e^{2\pi i j 2^{-1}} |k_1\rangle \otimes (e^{2\pi i j (k_2=0) 2^{-2}} |0\rangle + e^{2\pi i j (k_2=1) 2^{-2}} |1\rangle) \right) \\
 &= \frac{1}{\sqrt{2^2}} \left(\sum_{k_1=0}^1 e^{2\pi i j 2^{-1}} |k_1\rangle \otimes (|0\rangle + e^{2\pi i j 2^{-2}} |1\rangle) \right)
 \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{\sqrt{2^2}} \left((|0\rangle + e^{2\pi i j 2^{-1}} |1\rangle) \otimes (|0\rangle + e^{2\pi i j 2^{-2}} |1\rangle) \right) \\
&= \frac{1}{\sqrt{2^2}} \left((|0\rangle + e^{2\pi i \boxed{j} 2^{-1}} |1\rangle) \otimes (|0\rangle + e^{2\pi i \boxed{j} 2^{-2}} |1\rangle) \right)
\end{aligned}$$

$$j = J_1 J_2 \dots J_n = 2^n (0.J_1 J_2 \dots J_n)$$

Turn back to the case where $n = 2$.

$$j = J_1 J_2 = 2^2 (0.J_1 J_2)$$

$$j 2^{-1} = J_1 . J_2 \quad j 2^{-2} = 0 . J_1 J_2$$

Also note that $e^{2\pi i} = 1$ or $e^{2\pi i k} = 1$
 k is an integer.

Therefore ;

$$e^{2\pi i j 2^{-1}} = e^{2\pi i J_1 \cdot J_2} = \underbrace{e^{2\pi i J_1}}_1 e^{2\pi i 0 \cdot J_2} = e^{2\pi i 0 \cdot J_2}$$

$$QFT|j\rangle = = \frac{1}{\sqrt{2^2}} \left((|0\rangle + e^{2\pi i j 2^{-1}} |1\rangle) \otimes (|0\rangle + e^{2\pi i j 2^{-2}} |1\rangle) \right)$$

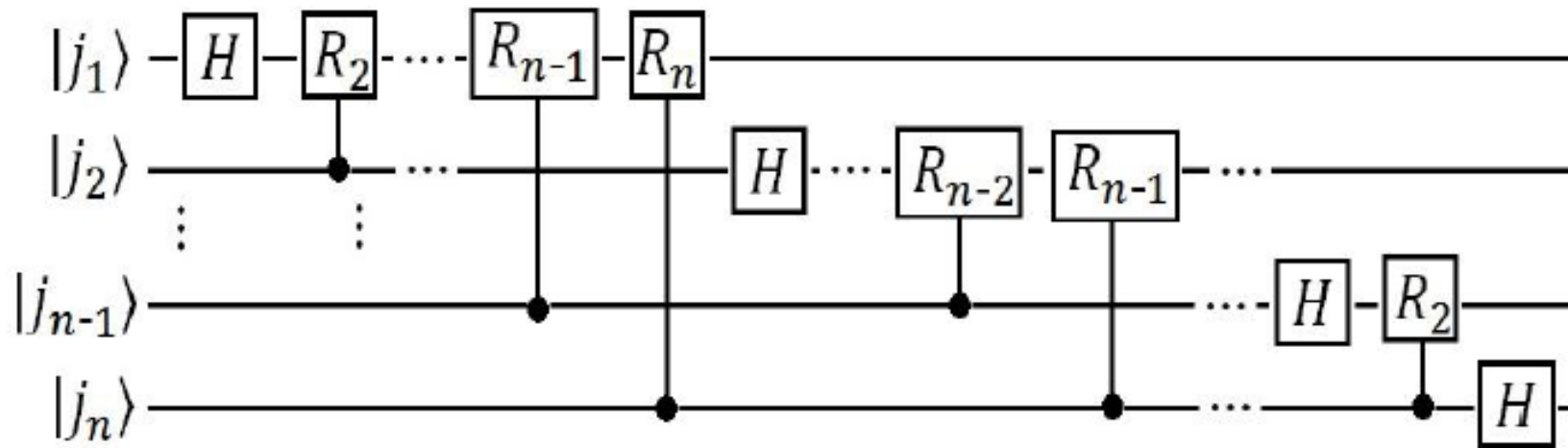
$$QFT|j\rangle = = \frac{1}{\sqrt{2^2}} \left((|0\rangle + e^{2\pi i 0 \cdot j_2} |1\rangle) \otimes (|0\rangle + e^{2\pi i 0 \cdot j_1 j_2} |1\rangle) \right)$$

Finally in the most general terms,

$$\begin{aligned}
 \text{QFT}_{2^n} |j\rangle &= \frac{1}{\sqrt{2^n}} (|0\rangle + e^{2\pi i 0 \cdot j_n} |1\rangle) \otimes (|0\rangle + e^{2\pi i 0 \cdot j_{n-1} j_n} |1\rangle) \otimes \\
 &\quad \dots \otimes (|0\rangle + e^{2\pi i 0 \cdot j_1 j_2 \dots j_n} |1\rangle) \quad (47) \\
 &= \frac{1}{\sqrt{2^n}} \bigotimes_{\ell=1}^n \left(\sum_{k_\ell=0}^1 e^{2\pi i j k_\ell 2^{-\ell}} |k_\ell\rangle \right)
 \end{aligned}$$

The output of QFT is an unentangled quantum product state.

How can we construct the QFT algorithm ?



when the state $|j_1 \dots j_n\rangle$ is input, Applying the Hadamard gate to the first bit produces the state

$$\frac{1}{2^{1/2}} \left(|0\rangle + e^{2\pi i 0 \cdot j_1} |1\rangle \right) |j_2 \dots j_n\rangle ,$$

$e^{2\pi i 0 \cdot j_1} = -1$ when $j_1 = 1$, and is $+1$ otherwise.

Applying the controlled- R_2 gate produces the state

$$\frac{1}{2^{1/2}} \left(|0\rangle + e^{2\pi i 0 \cdot j_1 j_2} |1\rangle \right) |j_2 \dots j_n\rangle .$$

We continue applying the controlled- R_3 , R_4 through R_n gates, each of which adds an extra bit to the phase of the co-efficient of the first $|1\rangle$. At the end of this procedure we have the state

$$\frac{1}{2^{1/2}} \left(|0\rangle + e^{2\pi i 0 \cdot j_1 j_2 \dots j_n} |1\rangle \right) |j_2 \dots j_n\rangle .$$

Next, we perform a similar procedure on the second qubit.
The Hadamard gate puts us in the state

$$\frac{1}{2^{2/2}} \left(|0\rangle + e^{2\pi i 0 \cdot j_1 j_2 \dots j_n} |1\rangle \right) \left(|0\rangle + e^{2\pi i 0 \cdot j_2} |1\rangle \right) |j_3 \dots j_n\rangle ,$$

and the controlled- R_2 through R_{n-1} gates yield the state

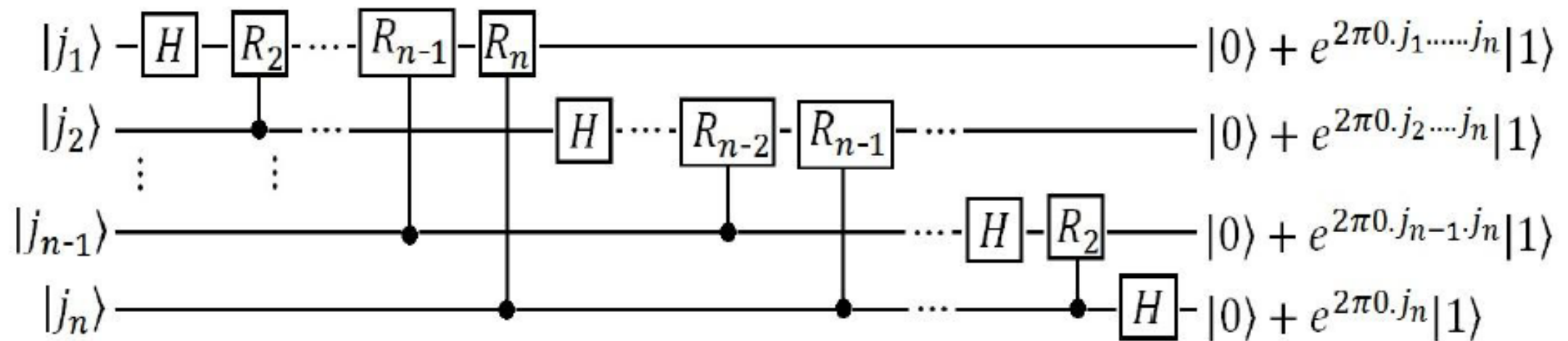
$$\frac{1}{2^{2/2}} \left(|0\rangle + e^{2\pi i 0 \cdot j_1 j_2 \dots j_n} |1\rangle \right) \left(|0\rangle + e^{2\pi i 0 \cdot j_2 \dots j_n} |1\rangle \right) |j_3 \dots j_n\rangle .$$

We continue in this fashion for each qubit, giving a final state

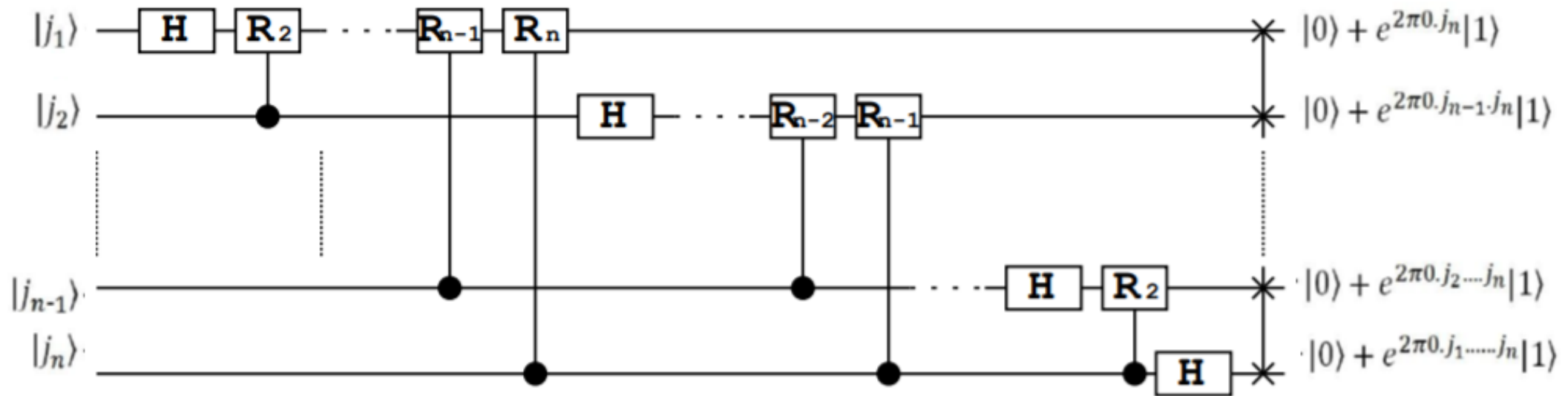
$$\frac{1}{2^{n/2}} \left(|0\rangle + e^{2\pi i 0 \cdot j_1 j_2 \dots j_n} |1\rangle \right) \left(|0\rangle + e^{2\pi i 0 \cdot j_2 \dots j_n} |1\rangle \right) \dots \left(|0\rangle + e^{2\pi i 0 \cdot j_n} |1\rangle \right) .$$

Is this the same result with Eq. (47) that we want to find?

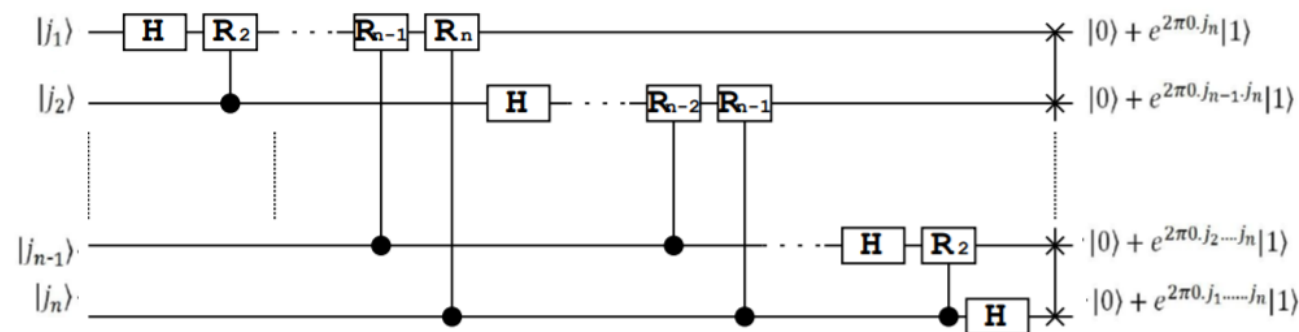
NO!



We need to reverse the order of the qubits at the end of the algorithm.



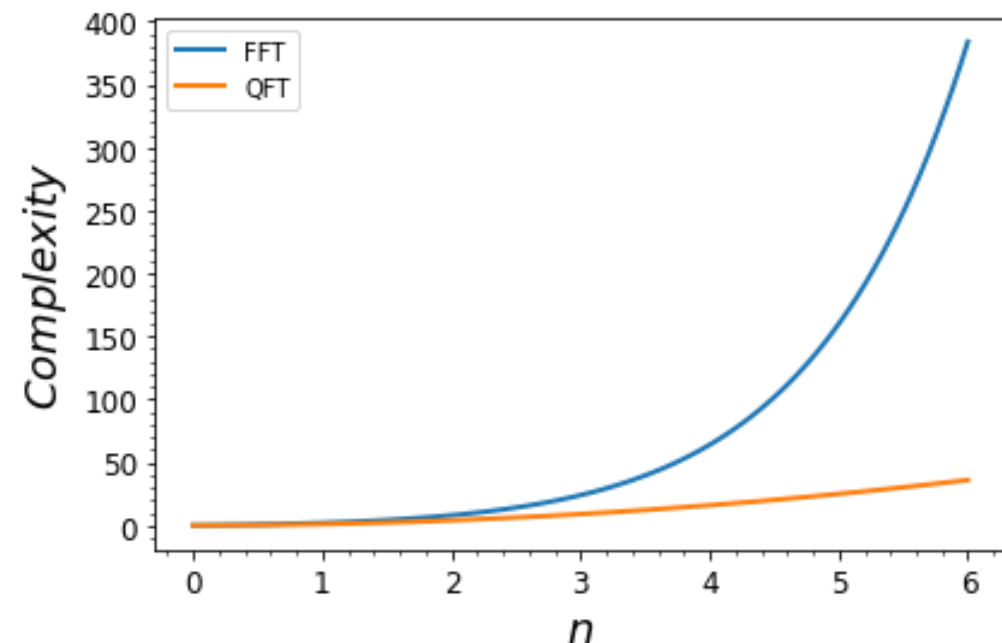
This is the final form of the n-qubit QFT algorithm.



Continuing in this way, we see that
 $n + (n - 1) + \dots + 1 = n(n + 1)/2$ gates are required,

Therefore, this circuit provides a $\Theta(n^2)$ algorithm for performing the quantum Fourier transform.

In contrast, the best classical algorithms for computing the discrete Fourier transform such as the *Fast Fourier Transform (FFT)*, compute using $\Theta(n2^n)$ gates.



inverse QFT circuit:

